

CREATIVE MISSIONS / TEACHER NOTES

HOW TO USE

Requests 31–42 extend the original 30 cards. Give each student one request plus the existing 01 common references and 02 operational worksheet. Keep teacher estimates and mission-specific incident updates separate until the initial estimate is submitted. All missions retain the measured four-worker, 20-minute production method, fill conversion and one-team/four-team/deadline comparisons. No initial crew, equipment or material allocation is supplied; required station counts are mission requirements.

ENGINEERING EXTENSIONS

Performance figures are explicit hypothetical assumptions, not equipment guarantees. Have students complete the original logistics calculations first. Use gallons = $\text{ft}^3 \times 7.48$; rainfall in feet = inches $\div$ 12. Basin capacity assumes the stated operating depth, not the crest. Report freeboard and overflow; add liner dimensions, corners and contingency where justified. The common reference's water-load estimate remains available; no sandbag weighing is required.

FIELD FORMAT

Keep the original supervised dry-site filling/closing/carrying/placement trial. These larger systems are paper missions. Optional tabletop demonstrations use shallow contained clean water and adult-operated apparatus. Qualified adults handle machinery and electrical systems; responders handle actual contamination. Real stream alterations and water-retaining structures require qualified review. Inspection and replacement are part of long-outage plans.

SOURCES AND ASSUMPTIONS

DOE explains that hydropower depends on flow and elevation drop and uses turbines/generators:

<https://www.energy.gov/cmei/water/how-hydropower-works>

NC State Extension describes ram pumps and the efficiency $\times$ source flow $\times$ fall/lift estimate:

<https://content.ces.ncsu.edu/hydraulic-ram-pumps>

EPA discusses containment/diversion, liners and overflow/underflow methods: <https://www.epa.gov/emergency-response-research/stormwater-decontamination-containment-and-diversion-technologies>

All mission dimensions, deadlines, loads, rainfall, machine ratings and incident changes were created for this exercise. Sources support the underlying functions, not these classroom layouts.

ESTIMATE 31 / LAST SIGNAL BEFORE DARK

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Illustrative narrow two-layer intake: $(12 + 6 + 6) \times 2 = 48$. Power = 39.24 W, nominally above 25 W; this does not guarantee charging reliability.

CORE CALCULATIONS

Illustrative $N = 48$ bags; fill = $48/75 = 0.64 \text{ yd}^3$; quarter-yard order = 0.75 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $48/S$; four-team time = $48/(4S)$. Usable hours = 3 minus delivery wait and separate setup/finish time. Minimum teams = round up $[48/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 1.07 hours for one team and 0.27 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

Leaves cut usable flow to 1 L/s. Recalculate power and decide which communications loads stay on.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 32 / THE WHEEL THAT KEEPS TURNING

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Intake: $16 \times 2 = 32$; channel edges: $30 \times 2 = 60$; total 92. Assumed delivery = 2.4 gal/min; 180 gallons requires 75 minutes after commissioning.

CORE CALCULATIONS

Illustrative $N = 92$ bags; fill = $92/75 = 1.23 \text{ yd}^3$; quarter-yard order = 1.25 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $92/S$; four-team time = $92/(4S)$. Usable hours = 4 minus delivery wait and separate setup/finish time. Minimum teams = round up $[92/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 2.04 hours for one team and 0.51 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

A branch jams the screen. Flow stops for 30 minutes; revise the watering schedule and inspection staffing.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 33 / WATER MUST CLIMB THE HILL

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Intake: $20 \times 2 = 40$ bags. Estimated delivery = 0.6 gal/min; four hours gives 144 gallons; 120 gallons takes 200 minutes.

CORE CALCULATIONS

Illustrative $N = 40$ bags; fill = $40/75 = 0.53 \text{ yd}^3$; quarter-yard order = 0.75 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $40/S$; four-team time = $40/(4S)$. Usable hours = 3 minus delivery wait and separate setup/finish time. Minimum teams = round up $[40/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 0.89 hours for one team and 0.22 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

Source flow falls to 3 gal/min. Recalculate supply and choose a backup or reduced watering target.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 34 / THIRTY PEOPLE, ONE WORKING TOILET

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Perimeter $40 \text{ ft} \times 4 = 160$. Capacity $= 24 \text{ ft}^3 = 179.52 \text{ gal}$. Forecast harvest $= 20 \text{ ft}^3 = 149.60 \text{ gal}$; demand $= 90 \text{ gal}$. Both quantities exceed demand under assumptions, but rainfall timing and transfer still matter.

CORE CALCULATIONS

Illustrative $N = 160$ bags; fill $= 160/75 = 2.13 \text{ yd}^3$; quarter-yard order $= 2.25 \text{ yd}^3$. Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time $= 160/S$; four-team time $= 160/(4S)$. Usable hours $= 2$ minus delivery wait and separate setup/finish time. Minimum teams $=$ round up $[160/(S \times \text{usable hours})]$; production workers $=$ teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 3.56 hours for one team and 0.89 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

Only 0.2 in of rain arrives. Recalculate harvest and the tanker quantity still needed.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 35 / CATCH THE RAIN BEFORE IT VANISHES

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Anchors = 36; basin = $28 \times 4 = 112$; total 148. Basin capacity 89.76 gal; forecast yield 62.832 gal, nominally above 50.

CORE CALCULATIONS

Illustrative $N = 148$ bags; fill = $148/75 = 1.97 \text{ yd}^3$; quarter-yard order = 2.00 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $148/S$; four-team time = $148/(4S)$. Usable hours = 3 minus delivery wait and separate setup/finish time. Minimum teams = round up $[148/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 3.29 hours for one team and 0.82 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

A tear reduces collection efficiency to 40%. Estimate the shortfall and plan repair time.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 36 / MUD IS KILLING THE PUMP

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Two perimeters: $2 \times 32 \times 4 = 256$; diversion = $12 \times 2 = 24$; total 280. Each cell holds 112.2 gal; volume $\div$ inflow = 37.4 min nominal residence time, not a sediment-removal guarantee.

CORE CALCULATIONS

Illustrative $N = 280$ bags; fill = $280/75 = 3.73 \text{ yd}^3$; quarter-yard order = 3.75 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $280/S$; four-team time = $280/(4S)$. Usable hours = 4 minus delivery wait and separate setup/finish time. Minimum teams = round up $[280/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 6.22 hours for one team and 1.56 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

One cell must be taken offline and inflow doubles. Recalculate nominal residence time and decide whether to pause transfer.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 37 / THE SLICK AT THE DRAIN

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Perimeter = 28 ft; $28 \times 4 = 112$ bags. Rated outlet exceeds inflow by 2 gal/min under the stated assumptions; no separation efficiency follows from that comparison.

CORE CALCULATIONS

Illustrative $N = 112$ bags; fill = $112/75 = 1.49$ yd³; quarter-yard order = 1.50 yd³. Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $112/S$; four-team time = $112/(4S)$. Usable hours = 2 minus delivery wait and separate setup/finish time. Minimum teams = round up $[112/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 2.49 hours for one team and 0.62 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

Debris reduces outlet capacity to 5 gal/min. Calculate net accumulation over 20 minutes; call for isolation/recovery rather than releasing the slick.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 38 / KEEP THE DIRTY WATER OUT

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Diversion: $52 \times 2 = 104$; collection pad: $36 \times 4 = 144$; total 248. Pad capacity = 149.6 gal; washwater = 120 gal, leaving 29.6 gal before the operating-depth limit.

CORE CALCULATIONS

Illustrative $N = 248$ bags; fill = $248/75 = 3.31 \text{ yd}^3$; quarter-yard order = 3.50 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $248/S$; four-team time = $248/(4S)$. Usable hours = 3 minus delivery wait and separate setup/finish time. Minimum teams = round up $[248/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 5.51 hours for one team and 1.38 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

The wash runs for 90 minutes instead. Calculate the recovery volume needed to prevent exceeding the operating-depth limit.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 39 / THE REPAIR WINDOW IS CLOSING

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Pocket = $28 \times 4 = 112$; apron anchors = $\text{ceiling}(24/4) = 6$; total 118. Net pumping = 4 gal/min; initial drawdown takes 10 minutes under constant assumptions.

CORE CALCULATIONS

Illustrative $N = 118$ bags; fill = $118/75 = 1.57 \text{ yd}^3$; quarter-yard order = 1.75 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $118/S$; four-team time = $118/(4S)$. Usable hours = 2 minus delivery wait and separate setup/finish time. Minimum teams = round up $[118/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 2.62 hours for one team and 0.66 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

Seepage rises to 7 gal/min. Explain why more bags or the original pump alone do not provide a dry repair window.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 40 / THE DARK-SKY AID STATION

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Tent 32 + solar 18 + cover 10 + diversion 80 = 140 bags. Students must allocate teams by dependency and check cart capacity over the longer route.

CORE CALCULATIONS

Illustrative $N = 140$ bags; fill = $140/75 = 1.87 \text{ yd}^3$; quarter-yard order = 2.00 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $140/S$; four-team time = $140/(4S)$. Usable hours = 2 minus delivery wait and separate setup/finish time. Minimum teams = round up $[140/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 3.11 hours for one team and 0.78 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

A thunderstorm warning ends outdoor operation. Reassign the plan to an indoor aid station and account for safe retrieval time.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 41 / THE LAST DRY SUPPLY LANE

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Diversion $40 \times 2 = 80$; path anchors ceiling $(56/4) = 14$; tray $20 \times 4 = 80$; total 174. Tray capacity 44.88 gal; captured muddy water needs removal.

CORE CALCULATIONS

Illustrative $N = 174$ bags; fill $= 174/75 = 2.32 \text{ yd}^3$; quarter-yard order $= 2.50 \text{ yd}^3$. Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time $= 174/S$; four-team time $= 174/(4S)$. Usable hours $= 2$ minus delivery wait and separate setup/finish time. Minimum teams $=$ round up $[174/(S \times \text{usable hours})]$; production workers $=$ teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 3.87 hours for one team and 0.97 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

The delivery truck cannot reach the planned unloading area. Filled bags and supplies now travel 450 ft; revise cart cycles, staffing and arrival time.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.

ESTIMATE 42 / FOUR CALLS, ONE CREW

POSSIBLE APPROACH • NOT A PRESCRIBED DESIGN

Intake $16 \times 2 = 32$; basin $32 \times 4 = 128$; diversion $36 \times 2 = 72$; tent 24; total 256. Basin capacity = 112.2 gal. Route-specific transport may dominate the schedule.

CORE CALCULATIONS

Illustrative $N = 256$ bags; fill = $256/75 = 3.41 \text{ yd}^3$; quarter-yard order = 3.50 yd^3 . Add justified corners, returns or contingency and recalculate if the layout changes.

R = measured completed bags in 20 minutes $\times 3$. Choose $S = 0.75R$ or $0.50R$, with justification. One-team time = $256/S$; four-team time = $256/(4S)$. Usable hours = 3 minus delivery wait and separate setup/finish time. Minimum teams = round up $[256/(S \times \text{usable hours})]$; production workers = teams $\times 4$. If usable time is zero or negative, change the plan.

Practice example only: $R = 60$, $S = 45$ gives 5.69 hours for one team and 1.42 hours for four teams. Actual staffing uses the student's rate and usable time.

RESOURCE AND SCHEDULE CHECK

Request empty bags, fill, delivery, two shovels per team as a starting point, gloves/glasses for every worker, appropriate transport and separately staffed support. Use the existing catalog's trips and route cycle times. Do not double-count movement already in the trial. Include technicians and inspections in dependencies, plus drinking water, rest, food when applicable and first aid. Additional bags cannot replace the listed machinery or qualified design.

INCIDENT UPDATE • ISSUE AFTER THE INITIAL PLAN

After your request is submitted, dispatch can deliver only 60% of your requested empty bags, rounded down, and half your requested production teams, rounded down. Prioritize, resize or relocate; calculate the revised service level.

ASSESSMENT

Require recalculated quantities and a revised schedule, service level and fallback. Accept justified alternative layouts. Keep engineering extensions distinct from the measured sandbag production rate.